

Q30 Ans: B

The corrected count rate initially is
 $320 - 20 = 300 \text{ Bq}$

The corrected count rate after 9 hours is

$$CCR = 300\left(\frac{1}{2}\right)^{\frac{9}{12}} = 178 \text{ Bq}$$

The observed count rate after 9 hours is

$$C = 178 + 20 = 198 \text{ Bq}$$

